



Automated oil palm fruit counting using deep learning for reliable yield estimation in natural background

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ABSTRACT

Accurate fruit counting is a critical component of yield estimation and harvest planning in oil palm plantations, yet manual counting remains labor-intensive and unreliable due to dense clustering, occlusion, and illumination variability under real field conditions. To address this challenge, this study introduces a density-aware benchmark dataset comprising 539 high-resolution RGB images of oil palm fresh fruit bunches (FFBs) collected in natural plantation environments in East Kalimantan, Indonesia. The dataset includes instance-level bounding-box annotations and captures substantial variability in fruit density, maturity stages, viewing angles, and lighting conditions. Three state-of-the-art deep learning architectures: YOLOv5, YOLOv8, and Mask R-CNN, were comparatively evaluated under a standardized image-level dataset split and common evaluation protocol. Beyond conventional detection metrics, including Precision, Recall, and mAP, a density-level counting analysis was conducted using Mean Absolute Error (MAE) across low-, medium-, and high-density fruit scenarios. Experimental results demonstrate that YOLOv5 achieved the most balanced overall performance, attaining a Precision of 0.963, Recall of 0.946, F1-score of 0.954, and the lowest global counting error, with an MAE of 4.96 fruits per image. YOLOv8 showed competitive detection capability with stable convergence behavior, while Mask R-CNN provided a complementary instance-based comparison but exhibited increased counting variability in high-density conditions. The findings confirm that lightweight one-stage detectors, particularly YOLOv5, offer a robust trade-off between detection accuracy and counting stability for practical deployment in large-scale oil palm plantations. The proposed density-aware benchmark and evaluation framework establish a reproducible foundation for scalable, AI-driven yield estimation in precision palm oil agriculture.

1. Introduction

Oil palm (*Elaeis guineensis* Jacq.) plays a pivotal role in the global vegetable oil industry, with Indonesia standing as the largest producer worldwide. The sector contributes substantially to national GDP and export revenues, reaching approximately 3.27% of Indonesia's GDP in 2019 [1,2]. Fresh fruit bunches (FFBs) serve as the primary raw material for crude palm oil, which is widely utilized in food, cosmetics, biodiesel, and detergent industries [3,4]. In this context, accurate estimation of fruit quantity per bunch is essential for yield forecasting, harvest scheduling, and efficient resource allocation in precision agriculture [5,6]. Unlike ground-level crops such as apples or strawberries, oil palm fruits grow in compact, high-altitude bunches, often obscured by fronds and overlapping structures [7,8]. These visual com-

plexities make manual counting and ground-based observation inefficient and unscalable for large plantations, limiting the integration of automation and data-driven decision-making in the palm oil sector [9,10].

Recent progress in computer vision and deep learning has enabled significant advances in automated fruit detection and counting across various crops [9,11]. However, research on oil palm imagery has predominantly focused on fruit ripeness classification [10,12,13] and disease detection [14,15], rather than on precise fruit counting. Furthermore, there is currently no publicly available benchmark dataset dedicated to oil palm fruit counting, and few existing studies evaluate models using counting-specific metrics such as Mean Absolute Error (MAE) alongside standard detection measures. This research gap is particularly critical because accurate detection does not necessarily translate to ac-

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curate counting. In dense bunches, overlapping and occluded fruits often lead to underestimation when using bounding box-based detectors, whereas instance segmentation approaches may offer higher precision at the expense of computational efficiency [7,16].

Developing an effective yield estimation framework therefore requires balancing detection accuracy, robustness to occlusion, and real-time inference capability especially for applications on edge devices such as drones and smartphones used in remote plantation monitoring [17]. To address these challenges, this study introduces a new benchmark dataset comprising 539 annotated RGB images of oil palm FFBs captured under real field conditions in East Borneo, Indonesia. The dataset covers diverse maturity stages, lighting variations, occlusion levels, and bunch orientations. Moreover, three state-of-the-art deep learning architectures YOLOv5, YOLOv8, and Mask R-CNN are comparatively evaluated under identical experimental settings to establish baseline performance for this domain. The primary focus of this study is therefore fruit detection and counting, while Mask R-CNN is included as a complementary instance-based architecture to examine the potential benefit of object separation capability in dense and occluded fruit regions.

This study bridges a critical gap between computer vision research and operational practices in the palm oil industry, laying the groundwork for scalable, automated yield estimation systems. The remainder of this paper is organized as follows: Section 2 reviews related work in fruit counting and oil palm vision systems; Section 3 details the dataset and methodology; Section 4 presents the experimental results and discussion; and Section 5 concludes the study with directions for future research.

2. Related works

The development of robust automated fruit counting systems is central to modern precision agriculture, serving as the primary mechanism for empirical yield forecasting and harvest optimization [18]. Unlike classification or detection alone, fruit counting requires resolving densely packed and partially occluded instances, which represents a technical challenge demanding both high recall and precise localization [29]. Recent studies have demonstrated the feasibility of deep learning for counting fruits in orchard environments. For example, Su et al. [19] achieved a MAE of 1.8 fruits per image for juvenile abalone counting using an improved SSD architecture, emphasizing the importance of integrating counting-specific metrics alongside standard detection benchmarks. In apple orchards, Maheswari et al. [16] compared DeepLabv3+ and YOLO variants and found that lightweight models like YOLOv8 offer the best trade-off between accuracy and inference speed for edge deployment. Similarly, Crespo et al. [7] reported that Mask R-CNN reduced strawberry counting error by 18% compared to bounding-box methods by leveraging pixel-level segmentation to disentangle overlapping fruits. These findings highlight a key dichotomy in agricultural counting, wherein detection-based approaches prioritize speed while segmentation-based methods favor accuracy in clustered scenes.

YOLO architectures have become dominant in real-time agricultural vision due to their single-stage design and efficiency. Nan et al. [11] introduced WGB-YOLO, a lightweight variant for pitaya detection, achieving 89% on mAP₅₀ at 42 FPS on edge GPUs. YOLOv8 improves gradient flow and multi-scale feature fusion via CSPDarknet and PANet, making it particularly suitable for small-object detection in cluttered agricultural scenes [20]. In contrast, Mask R-CNN is widely used as an instance segmentation architecture because it can generate object-level masks in addition to bounding boxes. In dense agricultural scenes, this capability may help separate overlapping objects, although its computational cost is generally higher than one-stage detectors [7]. Although computationally heavier, Mask R-CNN has been successfully adapted to agricultural contexts through optimization techniques such as TensorRT as in Crespo et al. [7] and model pruning as in Maheswari et al. [16]. These architectural advancements underscore the importance of selecting models that align with the specific challenges of dense fruit counting, particularly

in complex environments like oil palm plantations where occlusion and variability are prevalent.

Most computer vision research in oil palm has focused on ripeness classification and disease detection, not fruit counting [10]. Septiarini et al. [13] used contour-based segmentation to classify ripeness stages but did not count individual fruits. Josdaan et al. [17] employed YOLOv8 for ripeness detection with 92% accuracy, yet their dataset lacked instance-level annotations. For disease, Kent et al. [15] used UAV imagery and ResNet to detect basal stem rot, while Hamdani et al. [14] applied color histograms for leaf disease classification. However, none of these studies addressed quantitative yield estimation through direct fruit counting. Critically, no publicly available dataset exists for oil palm fruit counting. Existing resources such as datasets for oil palm disease or ripeness do not provide bounding boxes or masks at the fruit level. This gap limits reproducibility and benchmarking, hindering progress in automated yield estimation [5,6].

Recent literature emphasizes the need for multi-metric evaluation in agricultural computer vision. While mAP and Precision or Recall remain standard, studies increasingly incorporate counting accuracy (MAE) and inference efficiency (FPS) to assess real-world deployability [19]. For instance, Suhajito et al. [9] evaluated fruit detectors not only on detection performance but also on robustness to occlusion and lighting variation factors highly relevant to oil palm plantations. This evaluation framework aligns with the operational needs of plantation managers, who require systems that are not only accurate but also fast, lightweight, and robust under field conditions.

3. Material and methods

The methodological workflow of this study comprises five main stages: data acquisition, preprocessing and augmentation, ground-truth labeling, model development, and evaluation. The overall process follows a structured pipeline to ensure the reproducibility and robustness of the proposed fruit counting framework. The first stage, data acquisition, involved capturing high-resolution RGB images of oil palm fresh fruit bunches (FFBs) under natural field conditions. Images were collected from multiple viewpoints including top-down, side, and oblique angles to capture variations in bunch orientation, lighting, and occlusion. These diverse acquisition conditions represent the real-world complexity of plantation environments to ensure that the dataset is suitable for both controlled and operational settings. The workflow of the proposed methodology can be seen in Fig. 1.

In the preprocessing stage, each image was resized to 640 × 640 pixels to standardize the input size for model training. There was no sliding-window patch extraction or overlapping patch-based splitting was used in the final experimental pipeline. Various augmentation techniques were applied to enhance the model's ability to generalize across environmental conditions. These included random horizontal flipping, Contrast-Limited Adaptive Histogram Equalization (CLAHE), rotation, and random brightness-contrast adjustment. This augmentation step effectively simulates illumination differences, camera angle variations, and occlusion commonly encountered during data collection. The third stage, ground-truth generation, focused on manual annotation of visible individual fruit instances within each bunch. Human annotators marked individual fruits using bounding boxes, and the number of annotated boxes in each image was used as the reference fruit count. This dataset served as the foundation for training and validating the detection and counting models. Next, in the model development phase, the annotated dataset was split into training, validation/testing subsets.

The training data were used to optimize the model parameters, while the validation set guided early stopping and learning-rate adjustments. The model architecture consisted of a backbone convolutional network that extracted multiscale feature maps, followed by a region proposal network and fully connected layers for object classification, bounding box regression, and mask prediction. The framework was trained to predict the number of detected fruits per bunch by integrating both spatial

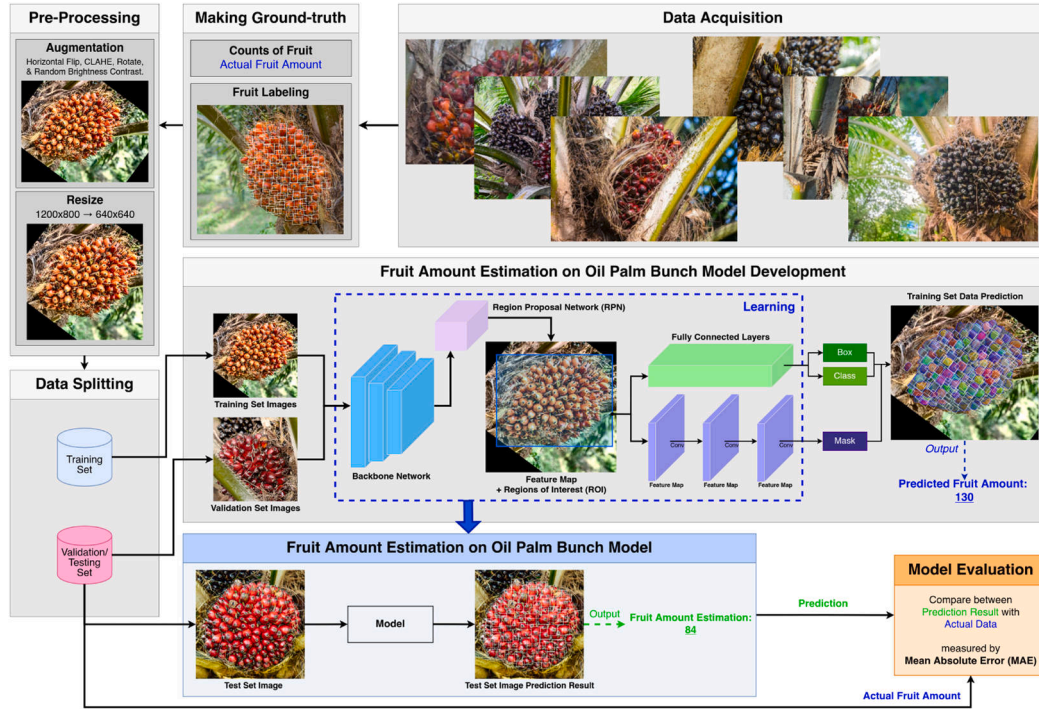


Fig. 1. Overall methodological workflow for automated oil palm fruit counting.

and contextual features within the image. Finally, the evaluation stage compared the predicted fruit count with the ground-truth annotations to assess estimation accuracy.

The counting performance was quantified using MAE, while detection precision, recall, and mean average precision (mAP) were computed following the COCO evaluation protocol. These quantitative evaluations were complemented by qualitative visualizations of predicted bounding boxes and detected instances to verify model reliability in dense and occluded fruit clusters. This structured workflow ensures end-to-end reproducibility from raw image acquisition to quantitative model evaluation and provides a scalable foundation for future development of automated yield estimation systems in oil palm plantations.

3.1. Data acquisition

The dataset was collected in Kutai Timur Regency, East Kalimantan, Indonesia (0.5607°N, 117.4245°E) during the early 2025 harvest season. A total of 539 RGB images of oil palm fresh fruit bunches (FFBs) were captured using a Xiaomi 12 Pro-smartphone (50 MP, f/1.9 aperture) under natural daylight, without artificial lighting or flash. Fruit bunches on-tree images were photographed from distances of 1.0–1.5 m and multiple angles (top-down, side, oblique) to maximize variability in occlusion and background.

The images exhibit diverse maturity stages (reddish orange to deep black), fruit densities, and levels of fiber occlusion, conditions representative of real plantation environments [13]. This variability is critical for developing models that generalize across operational scenarios Fig. 2 shows representative examples of oil palm bunches with different visual characteristics.

3.2. Dataset construction and annotation

Building upon the image dataset acquired during field observations in Section 2, a series of preprocessing and annotation steps were conducted to construct a reliable dataset for deep learning-based oil palm fruit detection and counting. The dataset consisted of 539 RGB images of oil palm fresh fruit bunches captured under natural field conditions

Table 1

Dataset distribution used in experimental pipeline.

Subset	Number of Images	Percentage	Purpose
Training	431	80%	Model Training
Validation/Testing	108	20%	Model Validation and Final Evaluation
Total	539	100%	Complete annotated dataset

with variations in illumination, bunch orientation, maturity stage, background complexity, and occlusion level (Table 1).

Manual annotation was performed using the LabelImg tool <https://github.com/HumanSignal/labelimg>. Each visible oil palm fruit instance was annotated using a bounding box and assigned the class label ‘oil_palm’. The annotations were initially stored in PASCAL VOC XML format and were subsequently converted into YOLO format to support training with YOLO-based detection models. Each image was paired with its corresponding annotation file to ensure consistency between the image data and label data throughout the experimental pipeline.

To avoid potential data leakage, the dataset was divided at the image level rather than at the object or patch level. Each original image and its corresponding annotation file were assigned exclusively to one subset. The dataset was split into 80% training data and 20% validation/testing data using a fixed random seed to ensure reproducibility. This strategy ensured that the same annotated image did not appear in more than one subset. The final experimental pipeline did not use overlapping patch-based splitting; therefore, the risk of data leakage caused by shared overlapping patches across training and validation/testing subsets was avoided.

The dataset was divided into 80% training data and 20% validation/testing data. Specifically, 431 original images were assigned to the training subset, while 108 original images were assigned to the validation/testing subset. The 20% subset was used for monitoring model performance during training and for reporting the final evaluation metrics.

The images were resized to 640 × 640 pixels to match the input requirements of the evaluated deep learning architectures. Data augmentation techniques were applied only to the training subset to reduce

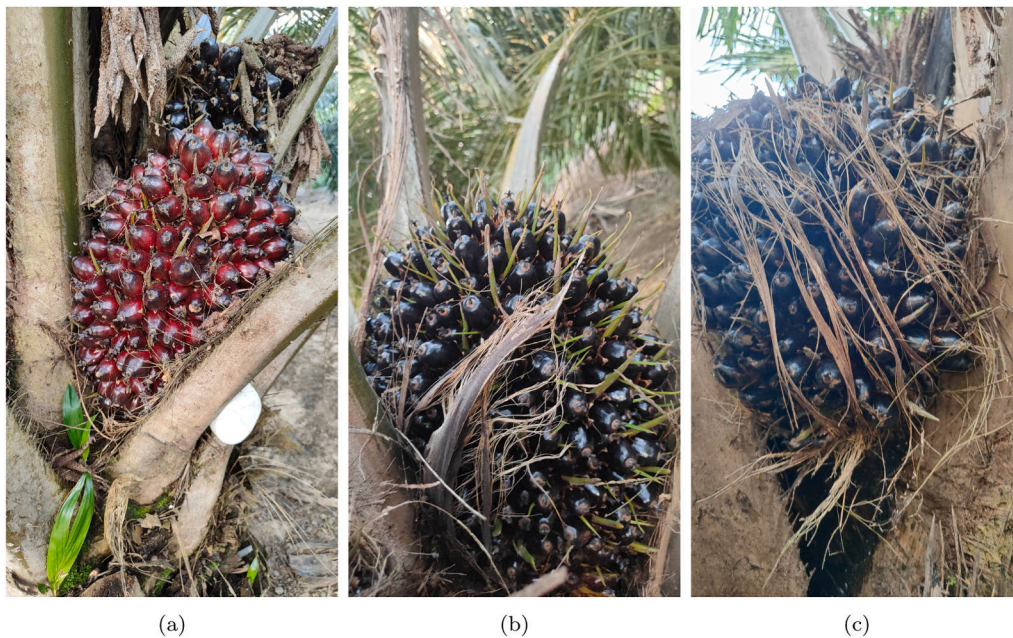


Fig. 2. Representative examples of oil palm fruit bunches in the dataset: (a) Ripe, minimal occlusion, (b) Unripe, moderate occlusion, and (c) Unripe, severe occlusion.

overfitting and improve model generalization under varying field conditions. The augmentation operations included horizontal and vertical flipping, brightness adjustment ($\pm 20\%$), Gaussian blur, and slight rotation. These transformations were selected to simulate real-world variations in illumination, camera angle, and partial occlusion commonly encountered in oil palm plantation environments.

The network weights were initialized using pre-trained parameters in each model to leverage transfer learning and accelerate convergence. During training, the model parameters were updated using the training subset, while the validation/testing subset was used to monitor model performance. A checkpoint mechanism also was applied by saving the model weights whenever validation mAP improved, to ensure that the best-performing model configuration was retained for final evaluation. The selected models were then evaluated using Precision, Recall, mAP₅₀, mAP_{50:95}, and Mean Absolute Error (MAE) to assess both detection and counting performance.

3.3. Models architecture

This subsection introduces the deep learning architectures evaluated in this study. Both one-stage object detectors and a two-stage instance-based model were selected to analyze the trade-offs between detection accuracy, counting robustness, and computational efficiency in dense fruit scenarios. Two state-of-the-art architectures were evaluated:

1. YOLOv5s [21] and YOLOv8n [20]: Single-stage object detectors optimized for speed and accuracy. Uses CSPDarknet backbone, PANet neck, and decoupled detection head. Selected for edge deployment feasibility.
2. Mask R-CNN (ResNet-50 + FPN) [22]: A two-stage instance-based model that performs object localization and can generate instance masks. Included as a complementary architecture to examine detection and counting behavior in dense and occluded fruit clusters.

The three architectures were selected to represent two different methodological families for oil palm fruit counting. YOLOv5s and YOLOv8n represent lightweight one-stage object detectors that are suitable for real-time deployment, while Mask R-CNN represents a two-stage instance segmentation approach that can provide pixel-level object separation in dense and occluded fruit clusters. Therefore, the comparison

designed as an architecture-level benchmark between detection-based and segmentation-based approaches.

This design allows the study to assess whether lightweight object detectors or a two-stage instance-based architecture are more suitable for practical oil palm fruit detection and counting under natural field conditions. Therefore, YOLOv5, YOLOv8, and Mask R-CNN were compared using common detection and counting metrics. Mask R-CNN was included to provide an instance-based comparison because its architecture can separate object instances in dense scenes, but the main reported outcomes remain detection accuracy and counting error.

All models were implemented using the PyTorch framework and trained on the same dataset split to ensure a fair comparison. The architectures were chosen based on their proven performance in agricultural vision tasks and their complementary strengths in detection and segmentation, providing a comprehensive evaluation of approaches for oil palm fruit counting under field conditions.

3.3.1. YOLOv5 and YOLOv8 architecture

YOLOv5 is a widely adopted single-stage object detection architecture known for its balance between accuracy and inference speed. It utilizes a CSPDarknet-based backbone for feature extraction, which enhances gradient flow while reducing computational redundancy. The neck incorporates a Path Aggregation Network (PAN) structure that facilitates multi-scale feature fusion, improving the model's ability to detect objects of varying sizes. The detection head employs a decoupled structure that separates classification and localization tasks, combined with distributional focal loss to enhance bounding box regression accuracy [21]. YOLOv5 was selected for its real-time detection capability, making it suitable for deployment on edge devices such as smartphones used in oil palm plantations. The overall network architecture of YOLOv5, including the backbone, neck, and detection head, is illustrated in Fig. 3.

YOLOv8 is a single-stage object detection architecture that balances accuracy and inference speed. The model adopts a CSPDarknet-based backbone for feature extraction, which enhances gradient flow while reducing computational redundancy. The neck integrates Feature Pyramid Network (FPN) and Path Aggregation Network (PAN) structures, enabling multi-scale feature fusion and strengthening the model's ability to detect objects of varying sizes. The detection head uses a decoupled structure that separates classification and localization tasks, combined

Overview of YOLOv5

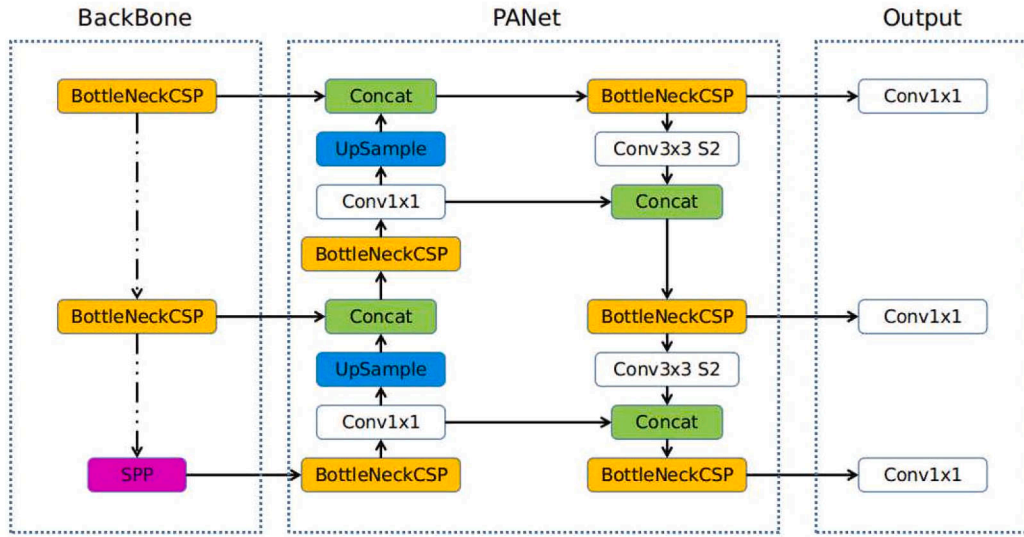


Fig. 3. YOLOv5 architecture [23].

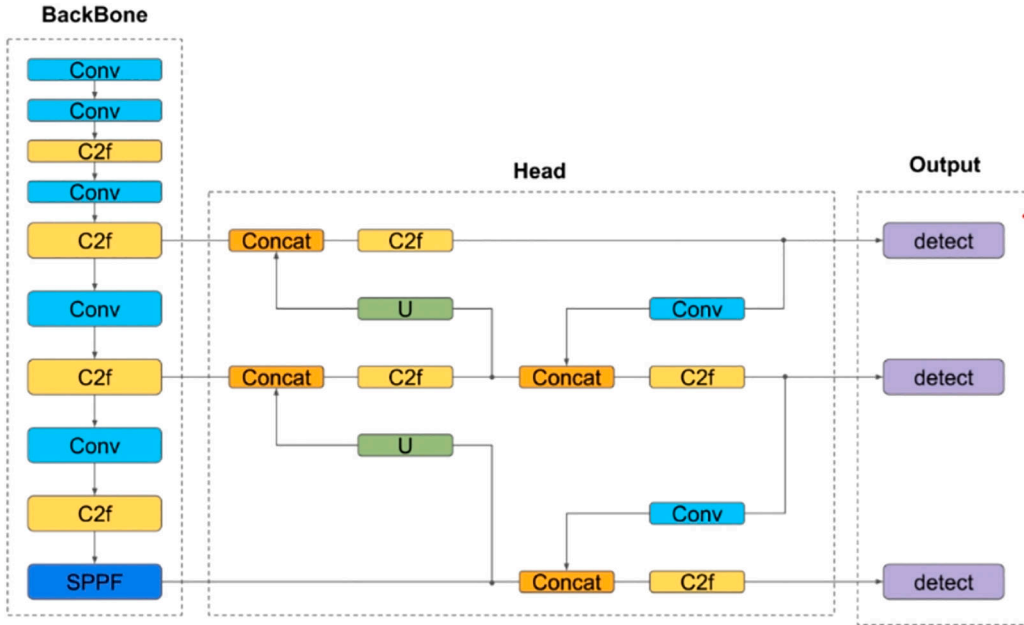


Fig. 4. YOLOv8 architecture [24].

with distributional focal loss to improve bounding box regression accuracy [20]. The overall network architecture of YOLOv8, including the backbone, neck, and detection head, is illustrated in Fig. 4.

To ensure efficiency for edge deployment, the YOLOv5s (small) and YOLOv8n (nano) variants were considered, as they provide a favorable trade-off between computational complexity and detection accuracy, making them suitable for mobile or resource-constrained environments such as oil palm plantations.

3.3.2. Mask R-CNN architecture

While bounding box detection is effective for most scenarios, occlusion and dense clustering in oil palm bunches may result in underestimation when fruits overlap. To examine this issue, Mask R-CNN was included as a complementary two-stage instance-based architecture. Mask R-CNN extends Faster R-CNN by adding an additional branch for in-

stance mask prediction while also producing object classification and bounding box localization outputs [22].

In the context of this study, Mask R-CNN was evaluated primarily using detection and counting metrics to maintain comparability with YOLOv5 and YOLOv8. Its inclusion was intended to assess whether an instance-based architecture could improve robustness in dense fruit regions. Therefore, the comparison focuses on Precision, Recall, F1-score, mAP where applicable, and MAE, rather than positioning this work as a standalone image segmentation study.

The segmentation masks provided by Mask R-CNN allow for a more precise delineation of individual fruits, even in cases where multiple fruits overlap within a bunch. Such capability is particularly valuable for oil palm fruit estimation, where bunches typically exhibit dense packing and partial occlusion by fibers. Previous agricultural studies have shown that Mask R-CNN achieves superior performance in tasks

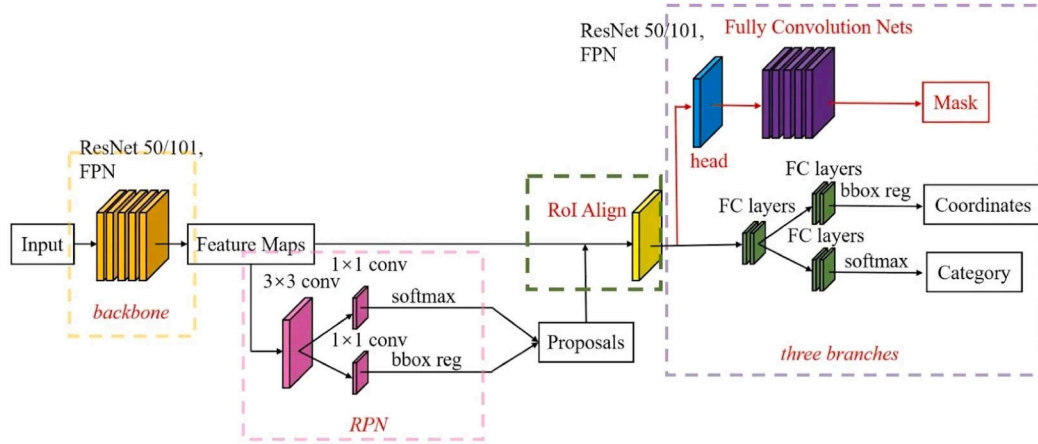


Fig. 5. Mask R-CNN architecture [25].

requiring fine-grained delineation, such as strawberry instance segmentation [7] and apple orchard fruit localization [16,26]. By adopting Mask R-CNN, this study evaluates the potential of segmentation-based methods to improve accuracy and robustness in oil palm fruit counting tasks. The architecture of Mask R-CNN, including the ResNet-50 backbone, Region Proposal Network (RPN) neck, and parallel branches for classification, bounding box regression, and mask prediction, is illustrated in Fig. 5.

3.3.3. Network training

The training of the deep learning models was performed using Google Colaboratory equipped with an NVIDIA A100 GPU (40 GB VRAM). The runtime environment utilized Python 3.10, PyTorch 2.0, CUDA 11.8, and cuDNN 8.7, ensuring compatibility with current deep learning frameworks. Model training and evaluation were executed within the Colab Pro+ environment, which provided approximately 51 GB of system RAM and 100 GB of temporary storage. This configuration enabled efficient parallel processing and reproducibility across multiple experimental runs, ensuring reproducibility and compatibility with modern deep learning frameworks.

The network weights were initialized from the pre-trained model to leverage transfer learning, as commonly practiced in agricultural object detection Nan et al. [11]. Images were resized to 640×640 pixels and trained for 100 epochs, following the standardized comparison protocol used in this study [3].

The optimizer was Stochastic Gradient Descent (SGD) with a momentum of 0.937 and a weight decay of $5e-4$, following best practices in YOLO optimization [20]. The learning rate was initialized at 0.01 and adjusted dynamically using a cosine annealing scheduler. Data augmentation techniques included random flipping, mosaic augmentation, brightness adjustments ($\hat{A} \pm 20\%$), and random rotations ($\hat{A} \pm 10^\circ$).

Early stopping was implemented with a patience of 10 epochs to prevent overfitting, and the best model weights were saved based on validation mAP. The same training protocol was applied to all three architectures to ensure a fair comparison of their performance under identical conditions. The hyperparameter settings for each model are summarized in Table 2.

The hyperparameter settings were selected based on architecture-specific recommendations, computational feasibility, and validation stability. For YOLOv5s and YOLOv8n, the input resolution of 640×640 was used because it is a commonly adopted resolution in YOLO-based object detection and provides a practical balance between preserving object details and maintaining computational efficiency. A batch size of 16 was selected based on GPU memory availability and stable convergence during training. The SGD optimizer with an initial learning rate of 0.01,

Table 2

Hyperparameter settings for YOLOv5/v8 and Mask R-CNN models.

Parameter	YOLOv5/v8	Mask R-CNN
Input size	640×640	640×640
Batch size	16	8
Epochs	100	100
Optimizer	SGD (mom = 0.937, wd = $5e-4$)	AdamW (lr = $2.5e-4$, wd = $1e-4$)
Learning rate	0.01	0.00025
Weight Decay	5×10^{-4}	10^{-4}
LR schedule	Cosine annealing	Linear warmup + step decay
Pretrained weights	MS COCO	ImageNet/COCO Backbone
Early stopping	Patience: 10 epochs	Patience: 10 epochs

momentum of 0.937, and weight decay of 5×10^{-4} was adopted following common YOLO training practice and to support stable optimization.

For Mask R-CNN, a smaller batch size of 8 was used because the two-stage detection and mask prediction process requires higher memory consumption than YOLO-based detectors. AdamW with an initial learning rate of 0.00025 and weight decay of 10^{-4} was selected to provide stable optimization for the segmentation-based architecture. The number of training epochs was set to 100 for all evaluated models to maintain comparable training duration across architectures. Early stopping with a patience of 10 epochs was applied to reduce the risk of overfitting, and the best model checkpoint was selected based on validation mAP. These settings were therefore chosen to balance training stability, computational feasibility, and fair comparison across different model architectures.

The architecture-specific hyperparameter settings were used due to YOLO-based detectors and Mask R-CNN differ substantially in architecture and memory requirements. The YOLO models were trained using SGD with a batch size of 16, while Mask R-CNN was trained using AdamW with a batch size of 8 due to its higher computational and memory demand. Although the optimizer and batch size differed across architectures, the comparison was controlled by using the same image-level dataset split, identical input size of 640×640 pixels, the same number of training epochs, pre-trained initialization, the same validation/testing subset, and identical evaluation metrics. Thus, the reported results should be interpreted as a practical benchmark under architecture-appropriate training configurations.

To ensure a fair and consistent evaluation across different detection architectures, a unified training and validation pipeline was employed in this study. The annotated oil palm fruit bunch dataset was divided into training and validation subsets using an 80:20 ratio, where the training set was used for model optimization and the validation set served as the primary reference for performance assessment. All models were evaluated under identical experimental conditions and were initialized

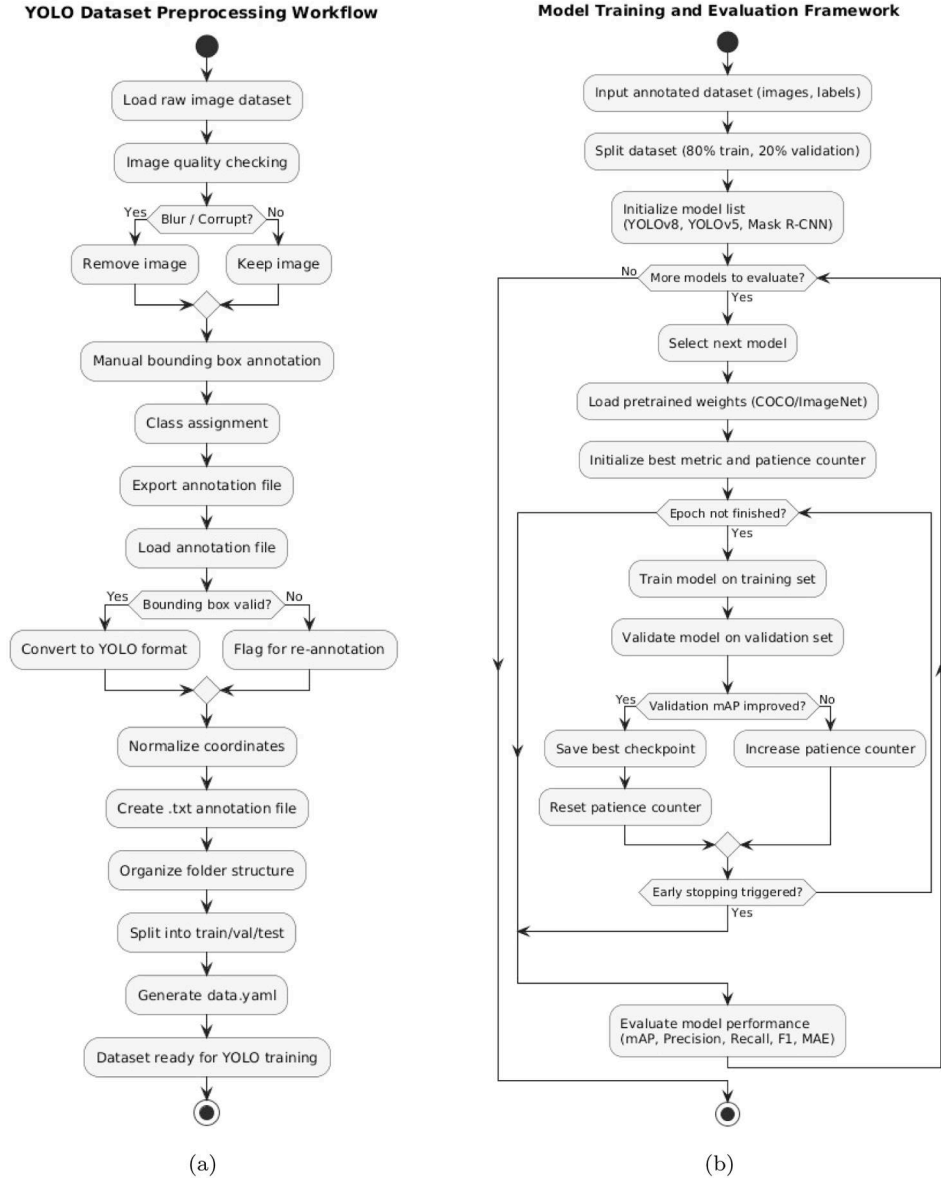


Fig. 6. Experiment workflows for palm oil fruitlet detection: (a) Dataset preprocessing (b) Training and evaluation pipeline.

using COCO pre-trained weights to accelerate convergence and improve feature generalization. During training, model parameters were iteratively updated on the training set, while performance was periodically monitored on the validation set using Precision and Recall as the main criteria.

Checkpoints were saved whenever an improvement in validation mAP was observed, ensuring that the best-performing model configuration was retained. This process was repeated until the predefined number of training epochs was completed. Finally, the optimal model for each architecture was selected based on its validation performance. The overall experimental workflow is illustrated in Fig. 6.

3.4. Standardized comparison protocol

To ensure a fair and reproducible comparison among the evaluated architectures, a standardized experimental protocol was applied. Although YOLOv5s, YOLOv8n, and Mask R-CNN have different architectural characteristics and architecture-specific training requirements, all models were evaluated using the same core experimental conditions. These included the same image-level dataset split, identical input resolu-

tion of 64×640 pixels, the same number of training epochs, pre-trained initialization, identical validation/testing subset, and the same evaluation metrics.

The comparison was designed as a practical architecture-level benchmark rather than a full hyperparameter ablation study. YOLOv5s and YOLOv8n were included to represent lightweight one-stage object detectors suitable for real-time deployment, whereas Mask R-CNN was included to represent a two-stage instance segmentation model that may provide advantages in dense and occluded fruit regions. Therefore, the standardized protocol focused on maintaining consistency in dataset usage, input configuration, training duration, and evaluation criteria while allowing each architecture to use its recommended optimization settings. The standardized comparison criteria are summarized in Table 3.

3.5. Performance evaluation

This subsection describes the evaluation protocol used to assess the detection and counting performance of the evaluated models. Since the primary objective of this study is oil palm fruit detection and counting, the evaluation focused on object detection accuracy and counting error.

Table 3
Standardized comparison setup for evaluated models.

Experimental aspect	YOLOv5s/YOLOv8n	Mask R-CNN
Dataset split	Same image-level split	Same image-level split
Input resolution	640 × 640	640 × 640
Training epochs	100	100
Pre-trained initialization	COCO	COCO / ImageNet backbone
Evaluation subset	Same validation / testing subset	Same validation / testing subset
Detection metrics	Precision, Recall, mAP	Precision, Recall, mAP
Counting metric	MAE	MAE
Comparison objective	Architecture-level benchmark	Architecture-level benchmark

Although Mask R-CNN can generate instance masks, it was evaluated using the same detection and counting metrics as YOLOv5 and YOLOv8 to ensure comparability across the evaluated architectures. Therefore, the quantitative comparison in this study focuses on detection accuracy and counting performance rather than standalone image segmentation evaluation.

Performance was assessed using two main groups of metrics: detection metrics and counting metrics. The detection metrics included Precision (P), Recall (R), mAP₅₀, and mAP_{50:95} based on the COCO evaluation protocol. The counting metric was Mean Absolute Error (MAE), which was used to measure the difference between the predicted fruit count and the manually annotated ground-truth count.

For object detection evaluation, the Intersection over Union (IoU) between predicted and ground-truth bounding boxes was computed. A detection was considered correct when the IoU exceeded a predefined threshold. The evaluation followed widely adopted object detection benchmarks, including the Pascal VOC and COCO protocols [27,28]. Precision (P), Recall (R), and Average Precision (AP), were computed as in Eqs. (1)–(4).

$$P = \frac{TP}{TP + FP} \quad (1)$$

$$R = \frac{TP}{TP + FN} \quad (2)$$

$$F1 = 2 \times \frac{P \times R}{P + R} \quad (3)$$

$$AP = \int_0^1 P(R) dR \quad (4)$$

Where TP, FP, and FN represent true positives, false positives, and false negatives, respectively. Precision measures the proportion of correctly detected fruit instances among all predicted detections, while Recall measures the proportion of correctly detected fruit instances among all ground-truth fruit annotations. Average Precision (AP) summarizes the precision-recall relationship across different confidence thresholds.

The overall detection performance was reported using mean Average Precision (mAP), including mAP₅₀ and mAP_{50:95}. The mAP₅₀ metric represents the average precision at an IoU threshold of 0.5, while mAP_{50:95} represents the average precision across IoU thresholds from 0.5 to 0.95 with an increment of 0.05. These metrics were used to evaluate both detection sensitivity and localization quality under different IoU thresholds.

For Mask R-CNN, the bounding-box detection outputs were used for quantitative comparison with YOLOv5 and YOLOv8. This approach was adopted because the main objective of this study is fruit detection and counting, not standalone image segmentation. Mask R-CNN was included as a complementary two-stage instance-based architecture to examine its detection and counting behavior in dense and occluded fruit regions. Thus, the reported quantitative comparison uses common detection and counting metrics across all evaluated models.

To assess fruit counting accuracy, Mean Absolute Error (MAE) was employed as the primary counting metric. MAE provides a direct measure of the average absolute difference between the predicted fruit count

and the manually annotated ground-truth count. It is defined as follows:

$$MAE = \frac{1}{N} \sum_{i=1}^N |CN_i - RQ_i| \quad (5)$$

Where N denotes the number of test samples, CN_i is the predicted fruit count for the i th sample, and RQ_i is the corresponding reference count obtained from manual annotation. According to Su et al. [19], a lower MAE indicates better counting accuracy and stronger agreement between the predicted and reference counts.

The predicted fruit count was obtained directly from the model outputs. For YOLOv5s and YOLOv8n, the number of fruits in each image was calculated by counting the valid bounding boxes remaining after confidence-threshold filtering and Non-Maximum Suppression (NMS). For Mask R-CNN, the predicted fruit count was calculated from the number of valid detected instances after confidence-based filtering of the instance outputs. No separate regression-based counting model or additional manual correction was applied. The MAE was then computed by comparing the predicted count from each model with the corresponding ground-truth count obtained from manual annotation.

To quantify the uncertainty of the counting performance, 95% confidence intervals (CIs) for MAE were estimated using bootstrap resampling over the validation/testing images. The absolute counting error for each image was first computed as the absolute difference between the predicted count and the manually annotated ground-truth count. Bootstrap samples were then generated by resampling the image-level errors with replacement, and the 2.5th and 97.5th percentiles of the bootstrap MAE distribution were used as the lower and upper bounds of the 95% CI.

In addition, pairwise statistical comparisons between models were conducted using the Wilcoxon signed-rank test because the counting errors were paired across the same image set and were not assumed to follow a normal distribution. A significance level of ($p < 0.05$) was used to determine whether the difference between two models was statistically significant.

4. Results and discussion

This section presents the qualitative and quantitative results of the evaluated models for oil palm fruit detection and counting. The discussion includes qualitative detection examples, quantitative detection performance, counting error analysis, precision-recall and F1-score comparison, confusion matrix analysis, contextual comparison with related studies, and discussion of study limitations and real-world applications.

4.1. Qualitative detection results

To provide an initial qualitative assessment of the detection models, representative examples from the training and validation datasets are presented. These examples illustrate the diversity of visual conditions encountered in the oil palm plantation environment, including variations in fruit density, illumination, and occlusion. The visualized samples also highlight the complexity of the detection task prior to quantitative evaluation.

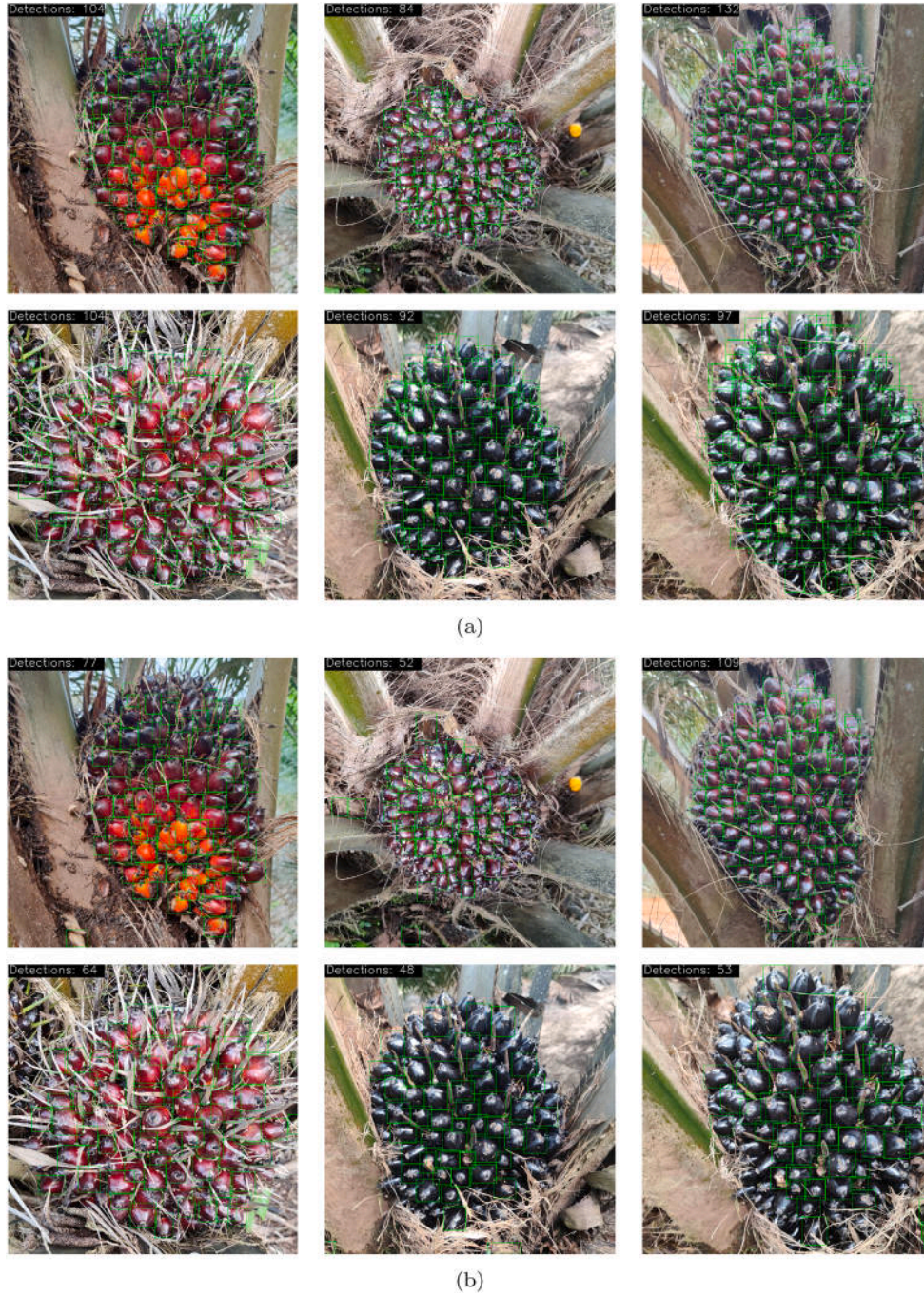


Fig. 7. Sample images from the training and validation sets with predicted bounding boxes: (a) YOLOv8n detections, (b) YOLOv5s detections.

Visual inspection of Fig. 7a and b reveals qualitative differences between the YOLOv5s and YOLOv8n detection models. Both models can identify oil palm fruit bunches under varying illumination conditions and background complexity. Their detection consistency however differs noticeably in dense and occluded scenarios. YOLOv8n results shown in Fig. 7a demonstrates more compact and stable bounding box predictions, particularly in images containing multiple overlapping fruit bunches. The predicted boxes tend to align closely with the ground-truth annotations, even when fruits are partially occluded by fronds or neighboring bunches. This behavior suggests improved feature representation and localization robustness in YOLOv8n.

In comparison, YOLOv5s detections shown in Fig. 7b produces reliable detections in moderately dense scenes but shows reduced consistency

when fruit bunches are heavily clustered. In such cases, YOLOv5s occasionally misses partially occluded instances or generates overlapping bounding boxes for adjacent fruits. These qualitative observations indicate that YOLOv5s adopts a more conservative detection strategy, which prioritizes precision but may lead to missed detections in complex scenes. Overall, the qualitative results are consistent with the quantitative performance reported in the subsequent sections, where YOLOv8n achieves higher recall and localization accuracy, while YOLOv5s maintains strong precision but exhibits slightly reduced sensitivity in densely populated fruit regions.

In addition to the YOLOv5 and YOLOv8 visualizations, Fig. 8 presents representative outputs of Mask R-CNN on validation images. The visualization shows that Mask R-CNN can detect oil palm fruit

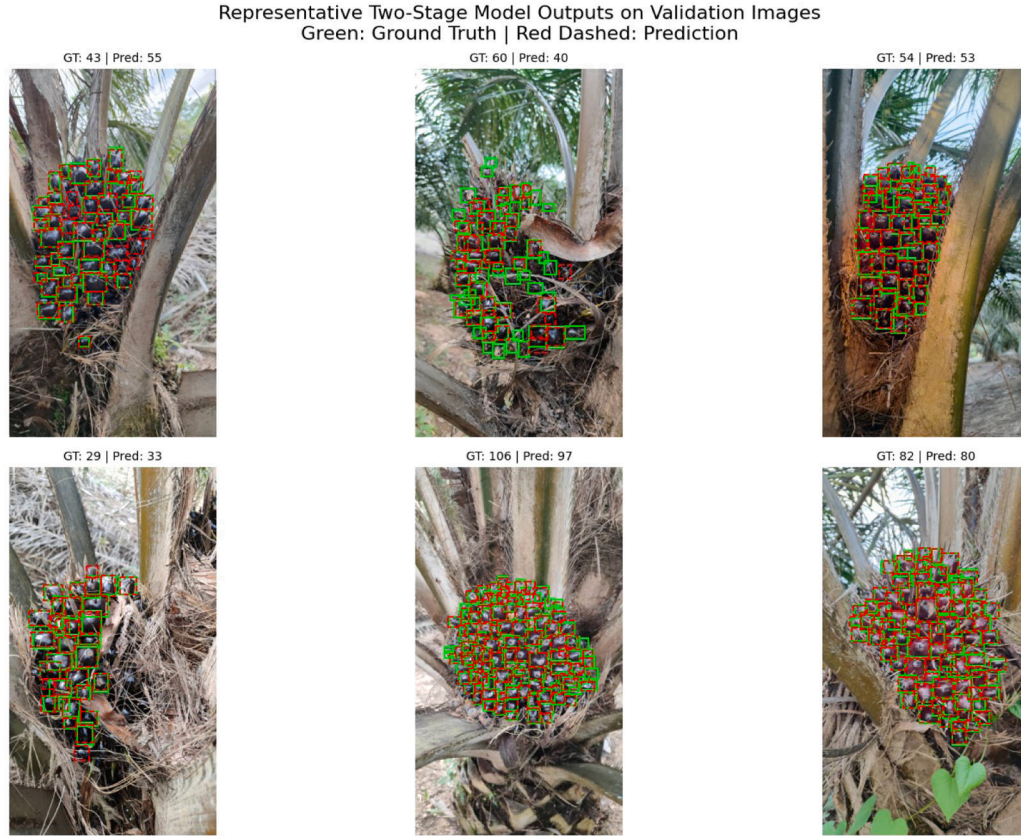


Fig. 8. Sample images from validation sets with Mask R-CNN predictions. Green boxes indicate ground-truth fruit annotations, while red dashed boxes indicate detected fruit instances. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 4

Quantitative comparison of model performance on the oil-palm fruit bunch dataset.

Model	Precision	Recall	mAP ₅₀	mAP _{50:95}
YOLOv5s	0.963	0.946	0.971	0.551
YOLOv8n	0.911	0.916	0.958	0.556
Mask R-CNN	0.925	0.913	—	—

instances under natural field conditions, including dense bunches and partially occluded fruit regions. In several examples, the predicted counts are close to the ground-truth counts, indicating that Mask R-CNN provides reliable instance-level detection outputs. However, some deviations are observed in visually complex scenes, particularly in dense bunches where fruits are highly clustered or partially hidden by fibers. These qualitative observations are consistent with the quantitative counting results, where Mask R-CNN remained competitive but exhibited higher counting variability compared with YOLOv5.

Table 4 presents the quantitative detection performance comparison among YOLOv8n, YOLOv5s, and Mask R-CNN on the validation dataset. Among the evaluated models, YOLOv5s achieved the highest detection accuracy, obtaining a Precision of 0.963, Recall of 0.946, and the highest mAP₅₀ of 0.971. These results indicate that YOLOv5s provides the most accurate localization and object detection performance with IoU threshold of 0.5.

YOLOv8n demonstrated comparable performance, with Precision of 0.911, Recall of 0.916, and mAP₅₀ of 0.958. Notably, YOLOv8n achieved a slightly higher mAP_{50:95} (0.556) compared to YOLOv5s (0.551), indicating stronger localization consistency under stricter IoU thresholds. Mask R-CNN achieved competitive Precision (0.925) and Recall (0.913), suggesting reliable detection capability. However, its overall detection

Table 5

Global MAE with 95% confidence intervals.

Model	Global MAE
YOLOv5s	4.96
YOLOv8n	6.87
Mask R-CNN	6.08

consistency under stricter IoU evaluation was lower compared to YOLO-based models.

This may be attributed to its two-stage detection mechanism, which emphasizes region proposal refinement and segmentation quality rather than maximizing bounding-box detection efficiency. Overall, the results demonstrate that one-stage YOLO detectors provide superior detection performance for oil palm fruit identification, while Mask R-CNN remains competitive and offers additional segmentation capability that may benefit boundary delineation tasks.

4.2. Counting performance analysis

To evaluate fruit counting accuracy, Mean Absolute Error (MAE) was used as the primary metric. Lower MAE values indicate closer agreement between predicted fruit counts and manually annotated ground truth counts.

As shown in Table 5, YOLOv8n achieved the lowest Global MAE, indicating the most accurate fruit counting performance across the validation dataset. YOLOv5s demonstrated comparable performance with slightly higher deviation. Mask R-CNN produced the highest MAE, suggesting increased counting errors, particularly in densely populated fruit clusters. The superior counting performance of YOLOv8n is consistent

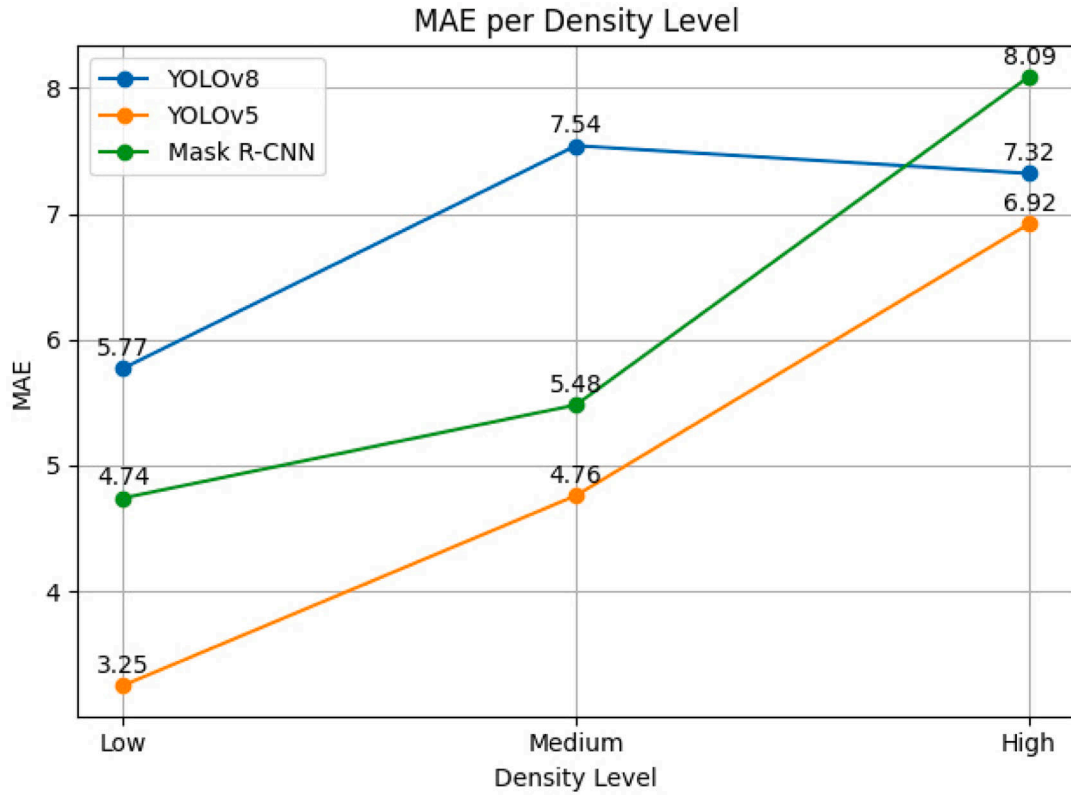


Fig. 9. MAE comparison across different fruit density levels.

with its higher Recall performance reported in Section 4.2. Since missed detections directly increase counting error, models with higher Recall tend to produce lower MAE values. To further analyze robustness under varying fruit densities, MAE was evaluated across three density levels: Low, Medium, and High.

As illustrated in Fig. 9, counting error increases with fruit density for all models. However, YOLOv8n maintains relatively stable performance even in high-density scenarios, demonstrating stronger robustness against occlusion and overlapping fruits. Mask R-CNN exhibits more pronounced error escalation under high-density conditions, which may be attributed to its lower Recall performance in complex scenes. To validate prediction consistency, a scatter analysis was conducted between predicted counts and ground truth annotations.

Fig. 10 present the scatter relationship between ground truth fruit counts and predicted counts for YOLOv8n, YOLOv5s, and Mask R-CNN, respectively. The red dashed diagonal line represents the ideal prediction scenario where predicted values perfectly match manual annotations. As shown in Fig. 10a, YOLOv8n demonstrates strong alignment with the ideal diagonal line, indicating high correlation between predicted and actual fruit counts. The distribution of data points remains tightly clustered around the diagonal, suggesting stable counting performance across varying fruit densities. In Fig. 10b, YOLOv5s exhibits similar behavior, with most predictions closely following the diagonal trend. However, slightly greater dispersion is observed in medium to high-count regions compared to YOLOv8n, reflecting minor counting deviation in denser fruit clusters.

In contrast, Fig. 10c shows that Mask R-CNN presents greater variability, particularly in high-count images. Several data points deviate below the diagonal line, indicating under-detection in densely clustered fruit areas. This observation is consistent with the lower Recall performance reported in Table 6 and the higher MAE values observed in Section 4.3. Overall, the scatter analysis confirms that YOLO-based one-stage detectors achieve stronger consistency with manual annotations in

Table 6

Precision, Recall, and F1-score comparison among models.

Model	Precision	Recall	F1-Score
YOLOv5s	0.963	0.946	0.954
YOLOv8n	0.911	0.916	0.913
Mask R-CNN	0.925	0.913	0.919

large-scale fruit counting scenarios, while Mask R-CNN provides complementary segmentation advantages despite slightly lower counting stability.

4.3. Precision-recall and F1-score analysis

To further examine detection reliability beyond mAP, Precision, Recall, and F1-score were analyzed. While mAP evaluates localization performance across IoU thresholds, Precision and Recall provide insight into the balance between false positives and false negatives. The F1-score, defined as the harmonic mean of Precision and Recall, summarizes overall detection stability.

As shown in Table 6, YOLOv5s achieved the highest Precision (0.963) and Recall (0.946), resulting in the highest F1-score (0.954). This indicates a well-balanced detection performance with minimal false positives and missed detections. YOLOv8n demonstrated slightly lower Precision and Recall values, yielding an F1-score of 0.913. Although marginally lower than YOLOv5s, its performance remains competitive and consistent with the mAP results reported in Section 4.2. Mask R-CNN achieved Precision of 0.925 and recall of 0.913, resulting in an F1-score of 0.919. While its F1-score slightly exceeds YOLOv8, its overall detection consistency under stricter IoU thresholds remains lower compared to YOLO-based models, as reflected in the mAP results. These findings confirm that YOLOv5s provides the most stable trade-off

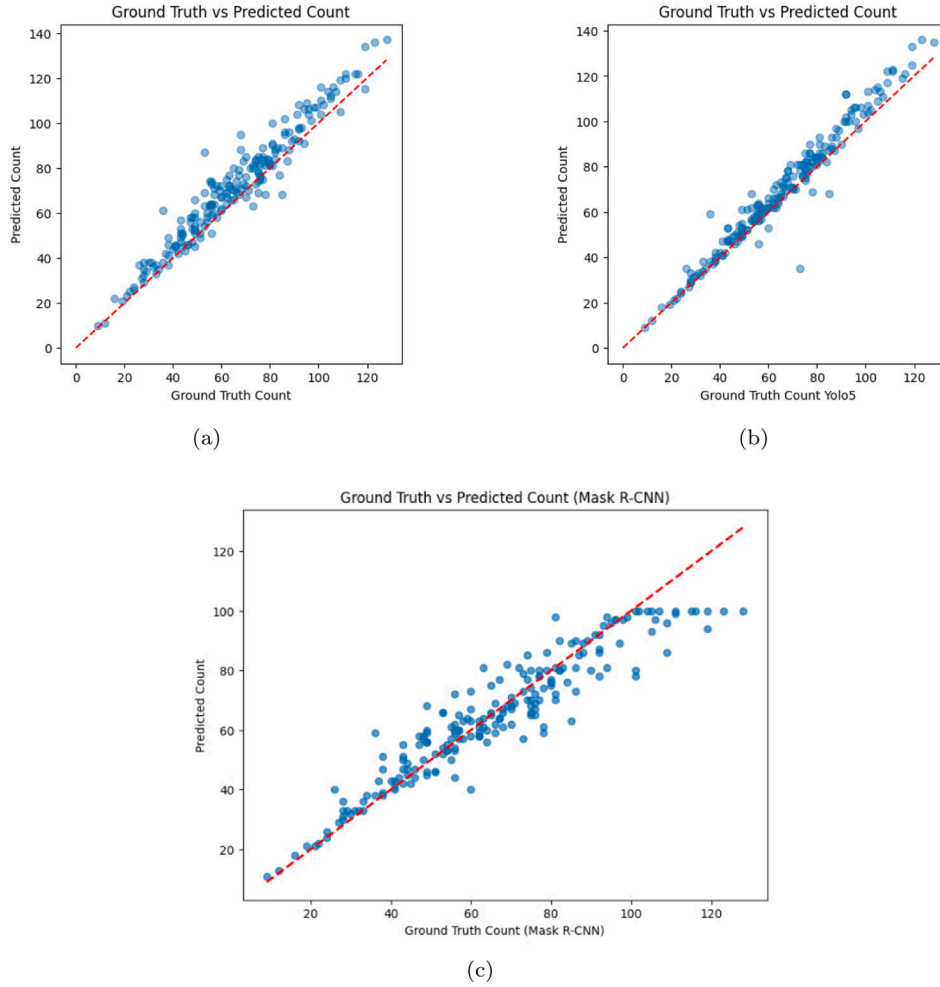


Fig. 10. Scatter plots comparing predicted fruit counts against ground truth annotations for (a) YOLOv8n, (b) YOLOv5s, and (c) Mask R-CNN.

between detection sensitivity and specificity. YOLOv8n maintains competitive performance, while Mask R-CNN demonstrates reliable but slightly less optimized detection balance for large-scale fruit detection tasks.

As illustrated in Fig. 11a and b, both YOLOv8n and YOLOv5s demonstrate stable convergence behavior throughout the training process. The training and validation loss curves decrease consistently over epochs, indicating effective optimization and minimal signs of overfitting. The close alignment between training and validation loss trends suggests that both models generalize well to unseen data.

The precision and recall curves show steady improvement as training progresses. YOLOv5s achieves slightly higher final precision and recall values compared to YOLOv8, reflecting stronger detection consistency. Similarly, the mAP_{50} and $mAP_{50:95}$ curves increase smoothly and plateau toward the later epochs, indicating that both models reach stable performance after sufficient training iterations. Notably, YOLOv5s exhibits marginally higher final mAP and more stable validation behavior, reinforcing its superior overall detection balance as previously observed in Table 6. These learning dynamics confirm that the reported quantitative metrics are supported by consistent training convergence patterns.

4.4. Confusion matrix analysis

To complement the quantitative evaluation metrics, a confusion matrix is presented to provide an intuitive view of the detection outcomes. In the context of object detection, confusion matrices are used in an illustrative manner to highlight the relationship between correctly detected

instances and missed detections, rather than to represent true negative background classifications. This visualization helps to better understand the detection reliability of the models on the validation set.

Confusion matrices illustrate the instance-level detection performance for the oil palm fruit class. As shown in Fig. 12a, YOLOv8n correctly detects approximately 95% of fruit instances, with around 5% missed detections. Fig. 12b indicates that YOLOv5s achieves a higher true positive rate of approximately 98%, reducing the missed detection rate to around 2%. This confirms the stronger recall performance of YOLOv5s observed in Table 6.

In contrast, Fig. 12c shows that Mask R-CNN correctly detects about 93% of fruit instances, with approximately 7% missed detections. Although the detection performance remains competitive, the slightly higher false detection and missed detection rates explain its lower F1-score compared to YOLOv5s. It is important to note that the absence of true negative values in the matrices reflects the instance-level nature of object detection tasks, where background regions are not explicitly treated as independent classification samples. Therefore, precision, recall, and F1-score remain the primary performance indicators for comparative evaluation. Overall, the quantitative results confirming that YOLOv5s achieves the most balanced detection performance among the evaluated models.

4.5. Comparison with state-of-the-art methods

Publicly available oil palm fruit counting benchmarks with instance-level annotations are still limited, a direct head-to-head comparison with

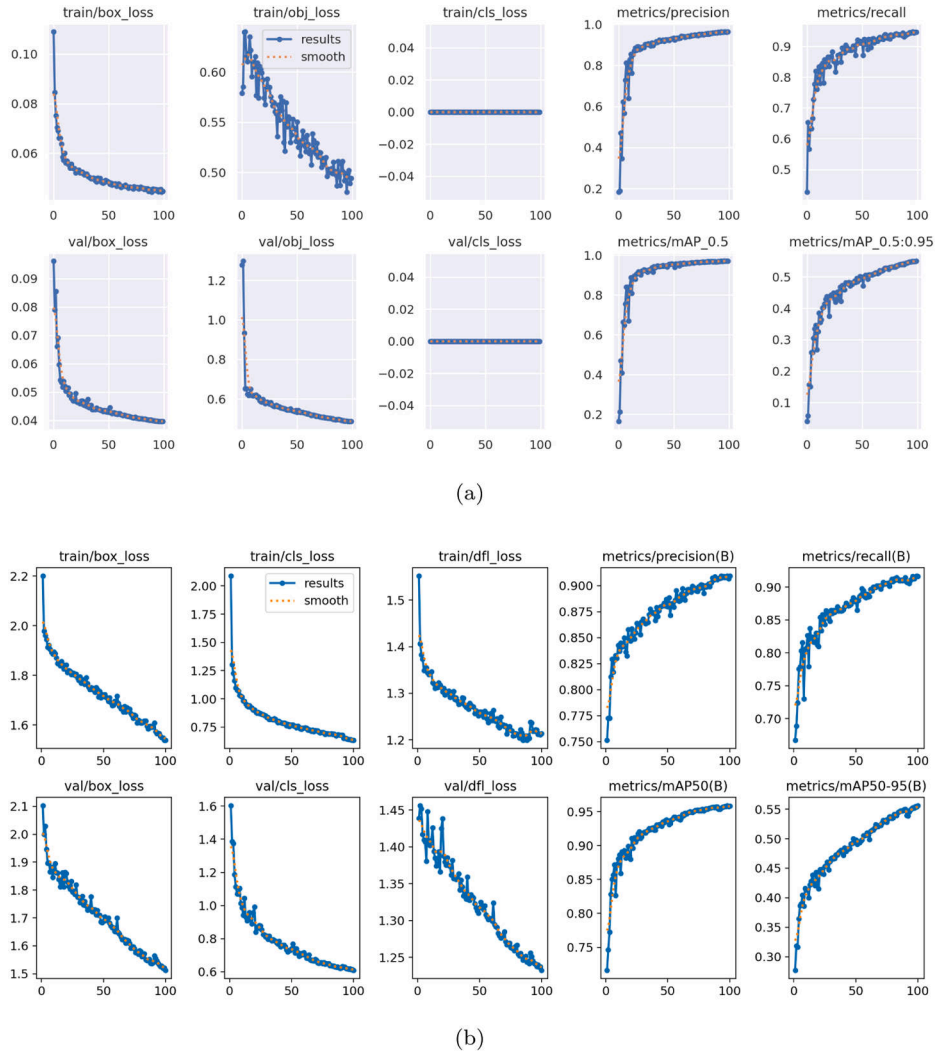


Fig. 11. Comparison of training loss, validation loss, and object detection metrics for (a) YOLOv8n and (b) YOLOv5s across training epochs, illustrating the convergence behavior and performance trends of both models during training.

previous methods on the same dataset is not feasible. Therefore, this study provides a contextual comparison with recent agricultural vision studies related to fruit detection, fruit counting, instance segmentation, and oil palm image analysis. The comparison is intended to position the contribution of the proposed dataset and evaluation framework, rather than to claim direct superiority across different datasets, crops, and acquisition conditions.

As shown in Table 7, previous studies have made important contributions to fruit detection, instance segmentation, and oil palm image analysis. However, most oil palm-related studies have focused on ripeness classification, disease detection, or FFB-level analysis rather than individual fruit counting. In contrast, the present study specifically addresses oil palm fruit counting under natural field conditions using instance-level annotations, MAE-based counting evaluation, confidence intervals, and density-aware analysis. Thus, the main contribution of this work lies in establishing a reproducible benchmark and evaluation baseline for oil palm fruit counting rather than in providing a direct cross-dataset performance ranking.

4.6. Limitations, future work, and real-world application

Although this study provides a benchmark dataset and comparative evaluation for oil palm fruit detection and counting under natural field conditions, several limitations should be acknowledged. First, the

dataset was collected from a specific plantation area in East Kalimantan, Indonesia. Although the images include variations in lighting, fruit density, maturity stage, viewing angle, and occlusion level, the dataset may not fully represent broader variations across different plantation regions, cultivars, seasonal conditions, camera devices, and management practices. Therefore, future work should expand the dataset across multiple geographic locations and acquisition periods to improve model generalization.

Second, the current evaluation was conducted using an image-level validation/testing subset derived from the available annotated dataset. While this setup provides a consistent basis for comparing YOLOv5, YOLOv8, and Mask R-CNN, a larger independent external test set would further strengthen the assessment of model robustness. Future studies should include cross-site validation or external testing using images collected from different plantations to better evaluate the transferability of the proposed approach.

Third, the counting results were derived directly from valid detection outputs after confidence filtering and Non-Maximum Suppression. This approach is practical and straightforward, but counting errors may still occur in dense bunches where fruits are highly overlapping, partially hidden by fibers, or visually merged with neighboring fruits. Future work may incorporate density-aware post-processing, instance refinement, or hybrid detection-segmentation strategies to reduce undercounting and overcounting in highly clustered regions.

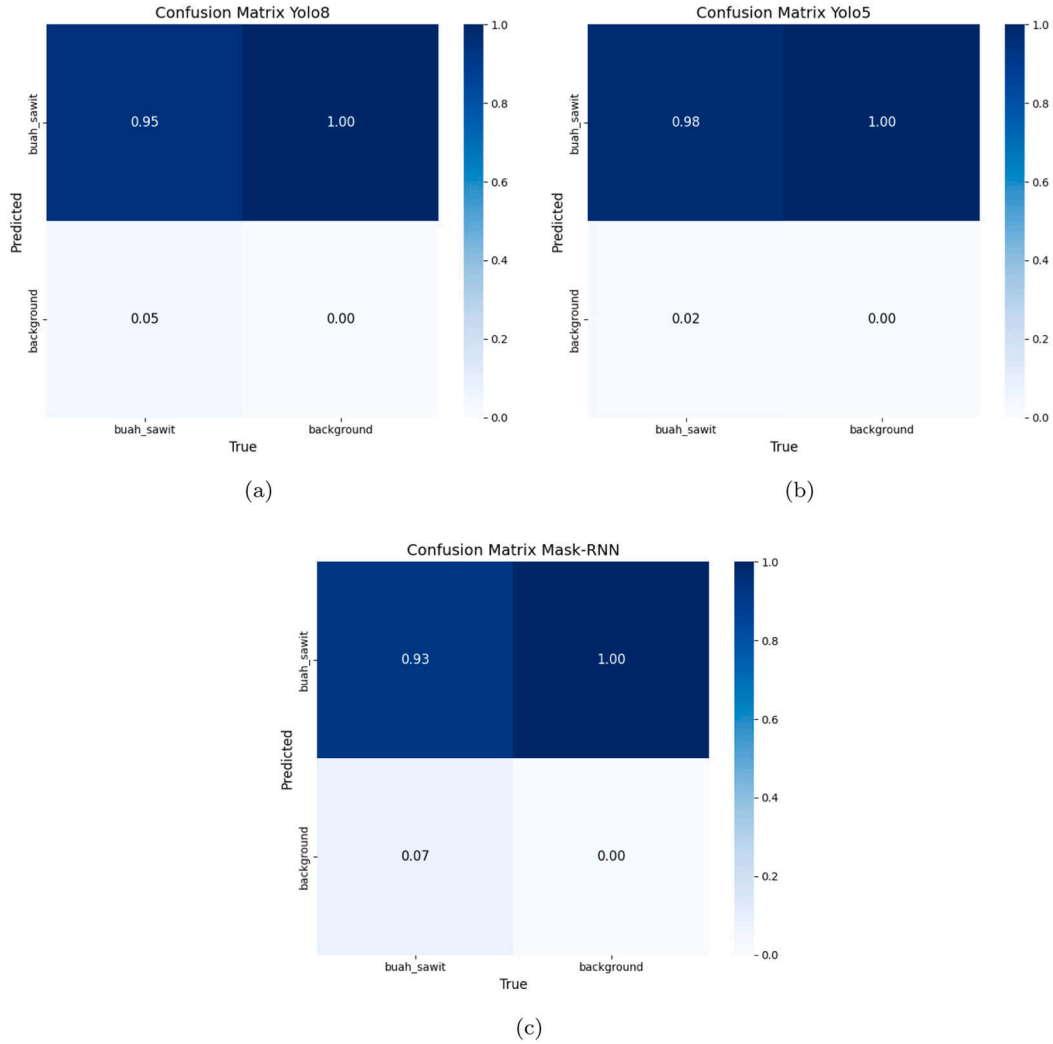


Fig. 12. Confusion matrices of (a) YOLOv8n, (b) YOLOv5s, and (c) Mask R-CNN models on the validation set.

Table 7

Contextual comparison with recent agricultural vision studies.

Study	Object	Method	Reported Metric	Relevance to This Study
Su et al. (2024) [19]	Juvenile abalone	Improved SSD	MAE = 1.8	Demonstrates the importance of MAE for object counting evaluation.
Nan et al. (2023) [11]	Pitaya fruit	WGB-YOLO	mAP@0.5 = 0.89; 42 FPS	Shows the effectiveness of lightweight YOLO-based models for fruit detection in natural agricultural environments.
Crespo et al. (2025) [7]	Strawberry	Mask R-CNN + TensorRT	Counting error reduced by 18%	Highlights the benefit of segmentation-based methods for overlapping fruit instances.
Maheswari et al. (2025) [16]	Apple	DeepLabv3+ and YOLO variants	Detection / localization performance	Compares detection and segmentation approaches for orchard fruit localization and deployment feasibility.
Josdaan et al. (2024) [17]	Oil palm FFB	YOLOv8	Accuracy = 92%	Applies YOLOv8 to oil palm imagery but focuses on ripeness detection rather than individual fruit counting.
Septiarini et al. (2020) [13]	Oil palm FFB	Image processing / contour-based segmentation	Classification / segmentation performance	Focuses on oil palm ripeness analysis and segmentation, not individual fruit counting.
Hamdani et al. (2021) [14]	Oil palm leaf	Color histogram and supervised classifier	Classification performance	Addresses oil palm health monitoring rather than fruit counting.
Kent et al. (2023) [15]	Oil palm tree	UAV imagery and deep learning	Disease detection performance	Demonstrates deep learning application in oil palm monitoring, but not fruit-level counting.
This study	Oil palm fruitlets / FFB	YOLOv5, YOLOv8, and Mask R-CNN	YOLOv5: Precision = 0.963; Recall = 0.946; F1-score = 0.954; MAE = 4.96; 95% CI = 4.33-5.66	Provides an oil palm fruit counting benchmark with instance-level annotations, MAE-based counting evaluation, confidence intervals, and density-aware analysis under natural field conditions.

Fourth, this study focused primarily on detection and counting accuracy using static RGB images. Although the results indicate that lightweight one-stage detectors, particularly YOLOv5, are promising for field deployment, real-time implementation on drones, smartphones, or embedded edge devices was not fully evaluated. Future work should investigate inference speed, memory usage, power consumption, and deployment optimization on mobile and edge platforms.

Finally, the current work provides an initial benchmark for automated oil palm fruit counting, but further integration with agronomic variables is needed for operational yield estimation. Future research should combine fruit count information with bunch weight estimation, maturity assessment, tree-level mapping, and temporal monitoring to develop a more comprehensive AI-driven yield forecasting system for precision oil palm agriculture.

Beyond these limitations, the proposed system has practical potential for supporting real-world oil palm plantation management. In operational settings, the model can assist plantation managers and field supervisors in estimating fruit abundance from field images, supporting harvest planning, labor allocation, and yield monitoring. Instead of relying only on manual visual estimation, users can capture oil palm bunch images using smartphones or drone cameras, and the system can automatically detect visible fruits and estimate their count. This can reduce manual counting effort, improve consistency across observers, and provide quantitative records for plantation monitoring. Smallholder farmers may also benefit from a low-cost decision-support tool for estimating production potential, while researchers and technology developers can use the benchmark dataset and evaluation framework as a foundation for developing more robust AI-based yield estimation systems.

5. Conclusion

This study presented a comprehensive evaluation of three modern deep learning architectures YOLOv5s, YOLOv8n, and Mask R-CNN for automated oil palm fruit detection and counting using a newly developed annotated dataset consisting of 539 field-acquired RGB images with instance-level labels. The dataset contributes to bridging the gap in agricultural computer vision research by providing structured annotations for dense fruit bunch detection. Experimental results demonstrate that YOLOv5s achieved the most balanced overall performance across detection and counting metrics. It obtained the highest Precision (0.963), Recall (0.946), and F1-score (0.954), while also exhibiting the lowest mean absolute error (MAE) across varying fruit density levels. YOLOv8n showed competitive performance with stable convergence behavior, though slightly lower recall compared to YOLOv5s. Mask R-CNN delivered reliable detection capability but demonstrated increased counting variability in high-density scenarios.

The findings confirm that lightweight one-stage detectors, particularly YOLOv5s, are highly suitable for practical oil palm fruit detection and counting applications, especially in large-scale plantation environments. Their balance between detection robustness and counting stability makes them strong candidates for field deployment using drone or mobile-based platforms. Future work will focus on expanding the dataset across multiple geographic regions, improving performance under severe occlusion conditions, and optimizing model deployment for embedded edge devices. This research establishes a solid foundation for scalable, AI-driven yield estimation systems in precision palm oil agriculture.

CRediT authorship contribution statement

Hamdani Hamdani: Writing – original draft, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization; **Anindita Septiarini:** Writing – review & editing, Supervision, Methodology,

Formal analysis, Conceptualization; **Encik Akhmad Syaifudin:** Writing – review & editing, Validation, Investigation, Formal analysis, Data curation; **Fazri Rahmad Nor Gading:** Writing – review & editing, Visualization, Software, Data curation; **Edy Winarno:** Writing – review & editing, Visualization, Validation, Formal analysis.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Anindita Septiarini reports financial support was provided by KEMENDIKTI SAINTEK INDONESIA. Reports a relationship with that includes: Has patent pending to. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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